C A P S T O N E P R O J E C T

Netflix Movies

Data Wrangling & Business Insights Report

1. Problem Statement

The streaming entertainment industry is one of the fastest-growing segments in media. Netflix, as a global leader, offers thousands of movies across diverse genres, ratings, and time periods. However, the sheer volume of content makes it difficult for stakeholders to understand which types of movies perform well, which genres dominate, and what content trends look like over time.

This project aims to clean, wrangle, and analyze a Netflix movies dataset to answer the following core business questions:

- Which genres are most prevalent on Netflix?
- How is the overall quality of Netflix content distributed by rating?
- What does the distribution of IMDb ratings look like across all titles?
- Is there a relationship between a movie's rating and its popularity (votes)?
- How has Netflix's content library grown year over year since 2000?

Answering these questions will help content strategists, analysts, and decision-makers make informed choices about acquisitions, marketing, and platform positioning.

2. Dataset Description

The dataset used for this project is n_movies.csv, which contains metadata on Netflix movie titles. Below is a summary of the dataset's key attributes:

File Name	n_movies.csv
Domain	Entertainment / Streaming Media
Minimum Rows	500+ (after filtering: movies from 2000 onwards with > 1,000 votes)
Source Type	Structured CSV (tabular data)

2.1 Key Columns

Column	Data Type	Description
title	String	Name of the movie
year	String → Integer	Release year (extracted from formatted string)
certificate	String	Age rating / content certificate (e.g., PG, R, U)
duration	String → Integer	Movie length in minutes
genre	String	Comma-separated genre list; first genre extracted as main_genre
rating	Float	IMDb user rating (0–10 scale)

votes	String → Integer	Number of IMDb user votes (comma-formatted in raw data)
rating_category	Derived String	Excellent (≥8), Good (6–8), Average (<6) — feature engineered
main_genre	Derived String	First genre extracted from the genre field
popularity	Derived Float	Rating × Votes — composite engagement score

3. Data Wrangling Steps

A systematic five-step wrangling process was applied to transform the raw CSV into a clean, analysis-ready dataset. Each step is documented below along with the rationale for the action taken.

3.1 Step-by-Step Summary

Action	Method & Rationale
Data Understanding	Loaded dataset with pd.read_csv(). Called df.info(), df.describe(), and df.isnull().sum() to understand structure, data types, and missing value counts before any transformation.
Handle Missing – Certificate	df['certificate'].fillna('Unknown') — preserves rows; unknown certificate is a valid, informative category.
Handle Missing – Genre	df['genre'].fillna('Unknown') — same rationale; genre is critical for analysis.
Handle Missing – Duration	Filled with mode (most common duration). Mode preferred over mean for duration as it avoids distortion from outliers.
Handle Missing – Rating	Filled with column mean. Rating is continuous; mean is the standard imputation for normally-distributed numeric features.
Handle Missing – Votes	Stripped commas, converted to numeric, then filled with median. Median is robust to the heavy right-skew typical in vote counts.
Drop Missing – Year	df.dropna(subset=['year']) — year is essential for temporal analysis; rows without it are unusable.
Correct Types – Year	Extracted 4-digit year from string using regex, then converted to integer.
Correct Types – Duration	Removed ' min' suffix, then converted to integer.
Feature Engineering – rating_category	Function mapped rating to Excellent (≥8), Good (6–8), Average (<6). Enables categorical analysis.
Feature Engineering – main_genre	Split genre string on ',' and extracted first element. Simplifies multi-genre entries for grouping.
Feature Engineering – popularity	Computed as rating × votes. A composite proxy for audience engagement.
Filtering – Year	Retained only movies released from 2000 onwards. Pre-2000 content is limited and outside modern streaming context.

Filtering – Votes	Removed titles with ≤ 1,000 votes. Very low-vote titles have unreliable ratings and skew analysis.

3.2 Why Each Step Matters

Data wrangling is not merely a technical necessity — it shapes the integrity of every downstream insight. Filling missing values prevents sample bias. Correcting data types ensures mathematical operations are valid. Feature engineering translates raw fields into business-meaningful signals. Filtering removes noise that would otherwise distort visualizations and conclusions.

4. Data Visualizations

Five visualizations were produced to explore different dimensions of the dataset. Each is described below in terms of its purpose, construction, and what it reveals.

4.1 Bar Chart — Top 10 Genres by Number of Titles

Chart Type: Horizontal Bar Chart

X-axis: Number of Titles | Y-axis: Genre (top 10 most frequent)

The chart uses a Blues colour gradient and horizontal bars for easy label readability. Each bar is annotated with its exact count. The chart is sorted in descending order so the dominant genres appear at the top.

Purpose: Identify which genres have the highest content volume on Netflix, helping content teams understand platform composition.

4.2 Pie Chart — Rating Category Distribution

Chart Type: Pie Chart with percentage labels

Segments: Excellent (≥8.0), Good (6.0–7.9), Average (<6.0)

A Set2 colour palette distinguishes the three quality tiers. Wedge edges are white for clarity. Percentage labels are formatted to one decimal place.

Purpose: Show the proportion of Netflix content that falls into each quality bracket, giving a quick sense of overall content quality.

4.3 Histogram — Distribution of IMDb Ratings

Chart Type: Histogram (30 bins)

X-axis: IMDb Rating | Y-axis: Number of Titles

Steel-blue bars with white edges reveal the shape of the rating distribution. This helps identify whether the Netflix library skews towards high-quality or mediocre content.

Purpose: Understand the spread and central tendency of IMDb ratings across all Netflix titles in the dataset.

4.4 Scatter Plot — Rating vs. Votes by Category

Chart Type: Scatter Plot (colour-coded by rating_category)

X-axis: IMDb Rating | Y-axis: Number of Votes

Three series (Excellent, Good, Average) are plotted with transparency (alpha=0.7) to show density. This chart captures whether higher-rated films also attract more votes.

Purpose: Explore the relationship between perceived quality (rating) and audience engagement (votes) across quality tiers.

4.5 Line Chart — Movies Released Over Time

Chart Type: Line Chart with markers

X-axis: Year | Y-axis: Number of Titles Released

Each year point is marked for precision. The chart covers titles released from 2000 onwards, revealing growth trends in Netflix's content addition strategy.

Purpose: Track how the volume of available Netflix movies has evolved over the past two decades.

5. Key Insights

The following insights were derived directly from the five visualizations. Each insight is paired with a business implication.

#	Visualization	Key Insight	Business Implication
1	**Bar Chart**	Drama and Comedy dominate the top genres, far outpacing other categories in title count.	Netflix should continue investing in Drama and Comedy originals, as these genres have the widest established audience base.
2	**Pie Chart**	The majority of Netflix content falls in the 'Good' (6–8) rating bracket. 'Excellent' content (≥8) is a minority, suggesting room for quality improvement.	Commissioning or acquiring more critically acclaimed titles could differentiate the platform and improve brand perception.
3	**Histogram**	IMDb ratings cluster between 6.0 and 8.0, forming a slightly right-skewed bell curve. Very few titles score below 5 or above 9.	The content library is largely mid-tier quality. A strategy to raise the floor (remove low-rated titles) or raise the ceiling (add prestige content) would sharpen positioning.
4	**Scatter Plot**	Excellent-rated movies tend to attract significantly more votes, indicating that quality and engagement are positively correlated.	High-quality content generates organic word-of-mouth. Investment in prestige content is likely to yield stronger user engagement and retention.
5	**Line Chart**	Netflix's title additions grew sharply post-2010, with a visible acceleration from 2015 onwards before potentially plateauing.	The rapid expansion phase may be slowing. A shift from quantity-based growth to a quality-focused curation strategy may be the next competitive move.

6. Business Recommendations

Based on the insights above, the following six recommendations are proposed for Netflix's content and business strategy:

Recommendation	Rationale
Double Down on Drama & Comedy	These are the top two genres by volume and likely by viewership. Expanding original productions in these genres maximizes reach.
Invest in Prestige / High-Rating Content	Excellent content (IMDb ≥8) correlates with more votes and engagement. Acquiring or producing award-calibre titles builds brand equity.
Curate and Retire Low-Performing Titles	Titles with very low ratings and few votes dilute catalogue quality. A periodic curation pass improves user experience.
Diversify into Niche Genres	While Drama and Comedy dominate, genres like Documentary, Animation, and Thriller may offer underserved growth opportunities.
Shift from Volume Growth to Quality Growth	The line chart shows content addition may be plateauing. Competing on quality rather than volume is the more sustainable long-term strategy.

7. Conclusion

This project demonstrates the end-to-end lifecycle of a data analytics capstone from raw, messy data to actionable business intelligence.

The Netflix movies dataset, after thorough wrangling, revealed a content library that is genre-concentrated, predominantly mid-tier in quality, and has experienced significant growth since 2000. The analysis shows that quality and audience engagement are correlated, and that Drama and Comedy are the platform's foundational content pillars.

Beyond the technical process, this project reinforced a key principle: data analysis is not about producing charts it is about telling a story with data. Every wrangling decision, every visualization choice, and every insight was made in service of answering real business questions that matter to real decision-makers.

The six business recommendations derived from this analysis provide Netflix's content strategy team with a clear, evidence-based direction: prioritize quality over quantity, invest in prestige content, and use engagement metrics not just view counts to guide curation.